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History of Astronomy: Greek Astronomy

From Myth to Science

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Introduction

Somewhere between the reign of the Assyrian kings and the rise of Rome, a handful of people living around the eastern end of the Mediterranean did something the world had never done before: they stopped telling stories about the sky and started building arguments about it. The Babylonians had watched the heavens longer, and recorded them more patiently, than anyone in history. The Egyptians had aligned their monuments to the stars with a care that still impresses surveyors. But neither culture asked the question that occurred to the Greeks: what must the sky be, such that it behaves as it does? That question, and the habit of answering it with geometry rather than legend, is the invention this book describes. We call the result Greek astronomy, and we live inside its consequences still.

The story runs from about 600 years before the common era to about 400 years after it, from Thales of Miletus, who according to tradition predicted a solar eclipse, to the later commentators who copied out Ptolemy's great treatise and wondered, in the margins, whether the wheels were really up there. In between stands the whole strange procession: Anaximander drawing the first map of the inhabited world and speculating about endless worlds; Pythagoras's followers deciding that the planets are a kind of music; Parmenides arguing the Earth into a sphere; Eudoxus hanging the planets on interlocking crystal spheres; Aristarchus proposing, two thousand years early, that the Earth goes around the Sun; Eratosthenes measuring the planet with a stick and a well; Hipparchus discovering that the sky itself drifts and writing down the brightness of the stars; and finally Ptolemy, assembling it all into the system that would command the world's attention for fourteen centuries.

Greek astronomy matters to the history of astronomy the way the alphabet matters to literature. It fixed the grammar that everything after would speak in: the sphere, the circle, the orbit, the table of positions, the catalogue of stars, the very idea that a heavenly body needs a mechanism rather than a myth. When Copernicus moved the Earth, he moved it inside a Greek machine; when Kepler found the ellipse, he found it by breaking Greek rules with Greek tools. Even our word for the wandering lights in the night, planet, is Greek for a wanderer, and our magnitude scale for star brightness descends directly from Hipparchus's six numbered classes.

One caveat belongs at the door, and the door is the right place for it. The cast of this story is almost entirely men whose works survive only in fragments and reports, and the standard history, this one included, is necessarily a reconstruction from partial evidence, in which a great deal of certainty attaches to the later tradition, above all to Ptolemy's complete treatise, and much less to the earlier centuries, where a single sentence of a lost book can carry a century of interpretation. Where the evidence is thin, this book says so; where the tradition is legend, it is labeled legend; and where the modern scholars still argue, the argument, not a false settlement, is reported. The reader should also know that the story is Greek only by language: the thinkers here came from Miletus and Samos, from Nicaea and Perga, from Alexandria and Rhodes, from the whole Greek-speaking world that stretched from Italy to Persia, and the astronomy they built was as international as the world that funded it.

This volume follows the series method: the discovery of the subject, the understanding of it, its science, its measurement, its context and its afterlife, and the questions it still leaves open. The reader should bring no special mathematics. The Greeks' arguments were geometric, and geometry ages better than algebra: most of them can still be seen whole, in the mind's eye, exactly as their inventors saw them, which is one of the quiet pleasures of this subject. What the reader should bring is the patience to watch a very human story unfold slowly: not the triumph of reason over superstition, which never happened, but something more interesting, the long argument among brilliant, eccentric, competitive men about how much the sky could be made to make sense.

Chapter 1: The Discovery of Greek Astronomy

Greek astronomy begins, like most Greek things, in a port town. Miletus, on the coast of what is now western Turkey, was one of the richest cities of the Greek world by the 500s before the common era, a hub of sea trade whose merchants sailed to Egypt, to the Black Sea, and into the rivers of the Near East. A merchant in Miletus dealt with Babylonian money-changers, Egyptian surveyors, and Phoenician pilots, and a thoughtful one could not help noticing that the foreigners who kept careful records of the sky were also the foreigners whose ships arrived and whose harvests held. The first Greek thinkers about nature, the men later called the Milesian philosophers, were of exactly this world: Thales, Anaximander, and Anaximenes were citizens of a commercial city whose business was navigation, and navigation, for a sailor, is applied astronomy.

Thales is the traditional starting point, and tradition has given him a biography full of legend: he is said to have

predicted the solar eclipse that is said to have stopped the armies of Lydia and Media in 585 before the common era, an event ancient writers dated by that very battle; he is said to have cornered the olive press market one spring to prove that a philosopher could be rich if he bothered; he is said to have fallen into a well while watching the stars, and to have been mocked by a servant girl for being eager to know the heavens and unable to see what was at his feet. Modern scholarship treats all of this gently. The eclipse prediction, if it happened, likely rested on Babylonian cycle lore the Greeks had only begun to learn; the anecdote about the olive presses is probably a moral tale polished by later generations. What survives the polishing is the essential claim, made by generations of Greek writers: with Thales, the sky entered the province of human calculation. An eclipse, until then the most terrifying of omens, could be anticipated. That single step, from omen to arithmetic, is the doorway this book walks through.

Anaximander, his younger fellow citizen, is in many ways the more striking figure, and far more of his ideas can be stated with confidence because ancient writers described them in detail. He taught that the Earth hangs motionless in the middle of everything, held there by nothing, because it is equidistant from every boundary of the cosmos; to modern ears the argument is circular, but notice what it assumes: that the cosmos has an order in which positions have reasons. He taught that the heavens are a set of rings, with the stars nearest, the Moon next, and the Sun farthest beyond, a scheme that gets the order wrong but the architecture right: a layered sky, each layer with a size and a distance. He is credited with drawing the first map of the known world, with introducing or popularizing the gnomon, the vertical rod whose shadow was Greece's principal astronomical instrument for the next thousand years, and with the first Greek prose book about nature, of which exactly one sentence survives. The Milesians also set the tone for the tradition's material culture, which was thin by later standards and inventive in spite of it. The ancients credited Thales with practical geometry learned in Egypt, where rope-stretchers re-established field boundaries after the Nile's floods, and with measuring the pyramids by their shadows, an anecdote that preserves a real link between surveying, shadow-clocks, and the astronomy to come. The instrument the tradition would use for its first five centuries, the gnomon, needed nothing but a straight stick and level ground; the calendar cycles needed nothing but tallies kept over generations; the observations needed nothing but clear skies and honest records. Greek astronomy began as a poor man's science, and some of its greatness lies in how far the poverty was made to reach: two cities' shadows measured the Earth, one old eclipse list measured the Moon, one eclipse that fell during a battle dated a whole dynasty's chronology.

A caution about the sources belongs here, at the tradition's beginning, because the beginning is where the sources are thinnest. Nothing survives from the Milesians' own hands; everything we know about them was written centuries later, by compilers and critics with their own axes, and the modern discipline of source criticism, which the field has practiced since the nineteenth century, spends much of its energy deciding which late reports deserve credit. The situation improves slowly with time: the fifth century before the common era is reported secondhand, the fourth has some direct quotation, the Hellenistic mathematicians are cited with increasing exactness, and only in the late tradition, Ptolemy above all, do we hold the books themselves. The reader should hold this gradient in mind through the early chapters: the stories of Thales and Anaximander are the tradition's own memory of itself, told by descendants, and the book treats them as memory, valuable and fallible, rather than as record.

Anaximenes, the third of the Milesians, kept the tradition of bold speculation, imagining the stars as nails in a crystalline vault and the Earth as a flat disc riding on air. All three were wrong about nearly everything, and that is not a criticism but the point: they were wrong in a new way, offering reasons a listener could examine and reject, rather than names of gods a listener could only honor.

Why this happened here, and not in older and richer civilizations, is one of the standing questions of the history of science, and the honest answer is that nobody knows. The standard guesses are worth listing, because each catches part of it. The Greeks were latecomers with no entrenched astronomical priesthood whose authority resisted change; they had absorbed, through trade, the recorded observations of Babylon and Egypt without inheriting the institutional habits that came with them; their alphabet had recently made writing common enough for arguments to circulate among private citizens; and their competitive civic culture prized winning arguments in public, which is very nearly a selective breeding program for people who make claims about the world. Whatever the mixture, the result was the discovery that nature could be an object of inquiry. Greek astronomy discovered itself, in this sense, before it discovered anything else: its first and greatest find was that the sky was findable.